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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech. (VLSI Design & Embedded Systems) (CBCS) Degree Examinations

May - June 2024

22VDE202 – REAL TIME OPERATING SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Discuss the service utility of the following real-time services and draw their service utility curves.
i) Hard real-time service. ii) Isochronal service. | 10 | L*2 | 1 | 3 |
| b) Write the pseudocode for a basic service that polls an input interface for a specific input vector. Modify the pseudocode for a basic event-driven software service. | 10 | L2 | 1 | 3 |
| 2. a) List and discuss the key features of an RTOS. | 10 | L2 | 1 | 3 |
| b) With neat diagram, discuss the thread states and their transition from one another. | 10 | L2 | 1 | 3 |

Unit – II

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|--|----|----|---|---|
| 3. a) Discuss the following priority policies with an example.
i) Earliest Deadline First. ii) Least laxity First. | 10 | L2 | 2 | 3 |
| b) Provide the mathematical proof for RM LUB using a simple two service case. | 10 | L3 | 2 | 3 |
| 4. a) Given three services S_1 , S_2 , and S_3 with service release periods $T_1 = 2$, $T_2 = 5$, and $T_3 = 7$, and service execution times $C_1 = 1$, $C_2 = 1$, and $C_3 = 2$. Draw schedule graph to find if they are schedulable under EDF and LLF. | 10 | L3 | 2 | 3 |
| b) Discuss the overload scenario of EDF scheduling policy. | 10 | L2 | 2 | 3 |

Unit – III

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|---|----|----|---|---|
| 5. a) Define WCET. List the conditions which guarantee full determinism in hard and soft real-time systems. | 10 | L2 | 3 | 3 |
| b) What is priority inversion? Explain. Briefly discuss the priority inheritance method. | 10 | L2 | 3 | 3 |
| 6. a) Discuss the system characteristics necessary for the deterministic execution time for a service. | 10 | L2 | 3 | 3 |
| b) Given the data word 11000100, perform hamming encoding for the following cases.
i) No errors in the received data. ii) Single bit error in bit-5. | 10 | L2 | 3 | 3 |

Unit – IV

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| 7. a) Differentiate between hardware breakpoints and software breakpoints in the pretext of debugging. | 10 | L2 | 4 | 3 |
| b) Explain in detail the following firmware components of an RTOS.
i) Boot code. ii) Device drivers. | 10 | L2 | 4 | 3 |
| 8. a) Discuss the following RTOS system software mechanisms:
i) Binary Semaphores. ii) Mutex Semaphores. | 10 | L2 | 4 | 3 |
| b) What are exceptions? Explain with simple example code how exceptions arise. Also explain how asserts help in handling exceptions with example codes. | 10 | L2 | 4 | 3 |

Unit – V

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|--|----|----|---|---|
| 9. a) Provide basic methods for optimizing code segments. | 10 | L2 | 5 | 3 |
| b) Discuss the design trade-offs between High Reliability and High Availability. | 10 | L2 | 5 | 3 |
| 10. a) Illustrate the effect of compiler generated assembly code on path length measurement using an example code in C language. | 10 | L3 | 5 | 3 |
| b) With a neat diagram, explain the concept of Drill-Down Tuning. | 10 | L2 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
